

Basic Statistics

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1 Statistical Inference

- Introduction
- Point Estimation
- Interval Estimation

Chapter 8: Statistical Inference

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- The main objective in any statistical enquiry is the properties of one or more population.
 - However, the population(s) is (are) usually unknown to us, and we simply have a sample from the population (or, a sample from each of the given populations)

- Statistical Inference:-

Given the properties of the sample (or, of the samples), to infer about those of the population(s) is the problem of statistical inference

- It is analogous to the inductive logic, the only difference being that the induction is achieved under probabilistic framework
 - Probability comes due to random sampling
- It is a process of going over from the known sample to unknown population.

Statistical Inference III

Statistical set-up of the problem of inference

- Let $(X_1, X_2, \dots, X_n)$ be a random sample of size n drawn from a population (discrete/continuous) with p.m.f/p.d.f $f(\underline{x}; \underline{\theta}) = f_{\underline{\theta}}(\underline{x})$, where $\underline{\theta}$ is the unknown parameter(s) of interest.
 - Our problem is to infer about $\underline{\theta}$
- Let Θ be the set of all possible values of θ
 - Θ is called the parameter space
- Note:
 - In the problem of statistical inference, Θ is known, although θ is unknown.
 - Example 1: $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$
 - $\theta = p$ is unknown, $\Theta = [0, 1]$ is known
 - Example 2: $X_1, X_2, \dots, X_n \sim \text{Normal}(\mu, \sigma)$
 - $\underline{\theta} = [\mu, \sigma]'$ is unknown, $\Theta = (-\infty, \infty) \times (0, \infty)$ is known
 - $\theta = \mu$ is unknown, $\Theta = (-\infty, \infty)$ is known
 - $\theta = \sigma$ is unknown, $\Theta = (0, \infty)$ is known

- Statistical Inference
 - ① Estimation
 - i Point Estimation
 - ii Interval Estimation
 - ② Hypothesis-testing

1 Estimation:-

Here, we have **no idea** about the true value of θ and the problem is **to estimate** the likely value of θ on the basis of the random sample $(X_1, X_2, \dots, X_n)$ drawn from the population

i Point Estimation:-

Here, we estimate θ by a **single** value (i.e., by a point)

- Let $T = T(X_1, X_2, \dots, X_n)$ be a statistic which is used to estimate the parameter θ , is called an **estimator** of θ
- For the observed sample $(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$, the observed value of the estimator, namely,

$$t = T(x_1, x_2, \dots, x_n)$$

is called an estimate of θ

ii Interval Estimation:-

Here, we estimate θ by an **interval** of values

- Let $T_1 = T_1(X_1, X_2, \dots, X_n)$ and $T_2 = T_2(X_1, X_2, \dots, X_n)$ be two statistics such that

$$P[T_1 \leq \theta \leq T_2] = 1 - \alpha,$$

where α is a pre-assigned small quantity. Usually, we take $\alpha = 0.05$ or 0.01 etc.

- If $\alpha = 0.05$, then $P[T_1 \leq \theta \leq T_2] = 0.95$. Hence, the observed values of $[T_1, T_2] = [t_1, t_2]$, say, is called a 95% confidence interval of θ

2 Hypothesis-testing:-

Here we have some idea about the true value of θ , in the form of a hypothesis, say, $\theta = \theta_0$,

- Our problem is **to judge or test** the validity/ feasibility/ tenability of the given hypothesis $\theta = \theta_0$ on the basis of random sample of the population

Chapter 8a: Point Estimation

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- In this case, we estimate θ by a **single** value (i.e., by a point)
 - Let $T = T(X_1, X_2, \dots, X_n)$ be a statistic which is used to estimate the parameter θ , is called an **estimator** of θ
 - For the observed sample $(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$, the observed value of the estimator, namely,

$$t = T(x_1, x_2, \dots, x_n)$$

is called an **estimate** of θ .

- Methods of calculating point estimates
 - Method of moments
 - Method of maximum likelihood

- Method of moments:

It consists in equating the first few moments of the population ($\mu'_k = E[X^k]$) with the corresponding moments of the sample

$$\left(m'_k = \frac{1}{n} \sum_{i=1}^n x_i^k \right), \text{ i.e.,}$$

$$\mu'_k = m'_k.$$

- The method of moments procedure:
Suppose there are l parameters to be estimated, say $\theta = (\theta_1, \dots, \theta_l)'$.
 - Find l population moments, μ'_k , for $k = 1, 2, \dots, l$.
 - μ'_k will contain one or more parameters $\theta_1, \dots, \theta_l$.
 - Find the corresponding l sample moments, m'_k , for $k = 1, 2, \dots, l$.
 - The number of sample moments should be equal to the number of parameters to be estimated.
 - From the system of equations, $\mu'_k = m'_k$, for $k = 1, 2, \dots, l$, solve for the parameter $\theta = (\theta_1, \dots, \theta_l)'$.
 - This will be a moment estimator of $\hat{\theta}$

Point Estimation (method of moment)

To Estimate	Notation	Point
Mean	$\bar{X}$	$\frac{1}{n} \sum_{i=1}^n X_i$
Proportion	$\hat{p}$	$\frac{1}{n} \sum_{i=1}^n I_{(X_i=1)}$
Variance	S_n^2	$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$

- Method of maximum likelihood:

It consists in choosing as estimator of $\underline{\theta}$ that statistic, which when substituted for $\underline{\theta}$, maximizes the likelihood function

$$L = f_{\underline{X}}(\underline{x}, \underline{\theta}) = f_{X_1, \dots, X_n}(x_1, \dots, x_n, \theta_1, \dots, \theta_l).$$

- Procedure to find maximum likelihood estimate (mle):

- Define the likelihood function, $L(\theta)$.
- Often it is easier to take the natural logarithm ($\ln$) of $L(\theta)$.
- When applicable, differentiate $l(\theta) = \ln L(\theta)$ with respect to θ , and then equate the derivative to zero.
- Solve for the parameter θ , and we will obtain $\hat{\theta}$.
- Check whether it is a maximizer or global maximizer.

- MLE for

- 1 Bernoulli: $\hat{p} = \bar{X}$

- 2 Exponential: $\hat{\lambda} = 1/\bar{X}$

- 3 Normal:

- 1 If μ is unknown and $\sigma^2 = \sigma_0^2$ is known, then $\hat{\mu} = \bar{X}$.

- 2 If $\mu = \mu_0$ is known and σ^2 is unknown, then $\hat{\sigma}^2 = S_n^2$

- 3 If μ and σ^2 are both unknown, then $\hat{\theta} = (\bar{X}, S_n^2)$.

Point Estimation VIII

- Invariance property of maximum likelihood estimators:

If $\hat{\theta}$ is the MLE of θ , then for any function $\tau(\theta)$, the MLE of $\tau(\theta)$ is $\tau(\hat{\theta})$.

Proof: Let us define

$$L^*(\eta|x) = \sup_{\theta: \tau(\theta)=\eta} L(\theta|x).$$

Let $\hat{\eta}$, denote the value that maximizes $L^*(\eta|x)$. We must show that $L^*(\hat{\eta}|x) = L^*(\tau(\hat{\theta})|x)$. Now, as stated above, the maxima of L and L^* coincide, so we have

$$\begin{aligned} L^*(\hat{\eta}|x) &= \sup_{\eta} \sup_{\theta: \tau(\theta)=\eta} L(\theta|x) \quad (\text{definition of } L^*) \\ &= \sup_{\theta} L(\theta|x) \quad (\text{iterated maximization is equal to the unconditional maximization over } \theta) \\ &= L(\hat{\theta}|x) \quad (\text{definition of } \hat{\theta}) \\ &= \sup_{\theta: \tau(\theta)=\tau(\hat{\theta})} L(\theta|x) \quad (\hat{\theta} \text{ is the MLE}) \\ &= L^*(\tau(\hat{\theta})|x) \quad (\text{definition of } L^*) \end{aligned}$$

Hence, $\tau(\hat{\theta})$ is the MLE of $\tau(\theta)$.

- Let $h(\theta)$ be a (not necessarily one-to-one) function of θ . If $\hat{\underline{\theta}} = (\hat{\theta}_1, \dots, \hat{\theta}_l)$ is the MLE of $\theta = (\theta_1, \dots, \theta_l)$, then the MLE of a function

$$h(\underline{\theta}) = (h(\theta_1), \dots, h(\theta_l))$$

is $h(\hat{\underline{\theta}}) = (h(\hat{\theta}_1), \dots, h(\hat{\theta}_l))$ for $1 \leq k \leq l$.

- MLE of σ is $\sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$ since it was shown that the MLE of σ^2 is $(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2)$.
- MLE of $\sqrt{p(1-p)}$, where p is a binomial probability, is given by $\sqrt{\hat{p}(1-\hat{p})}$.

Some Desirable Properties of Point Estimators

- Unbiased
- Sufficiency
- Consistency
- Efficiency

Unbiased Estimators

- A point estimator $\hat{\theta}$ is called an unbiased estimator of the parameter θ if

$$E(\hat{\theta}) = \theta$$

for all possible values of θ .

- Otherwise $\hat{\theta}$ is said to be biased.
 - Furthermore, the bias of $\hat{\theta}$ is given by

$$B(\hat{\theta}) = E[\hat{\theta}] - \theta.$$

- *Theorems:*

- The sample mean, $\bar{X} \left(= \frac{1}{n} \sum_{i=1}^n X_i \right)$ is an unbiased estimator of the population mean μ .

- Sketch of proof: $E[\bar{X}] = E \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \mu$

Unbiased Estimators III

- The statistic, $S_n^2 \left(= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \right)$ is not an unbiased estimator of the population variance σ^2 .

- Sketch of proof:

$$\begin{aligned} E[S_n^2] &= E\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right] = \frac{1}{n} E\left[\sum_{i=1}^n ((X_i - \mu) - (\bar{X} - \mu))^2\right] \\ &= \frac{1}{n} \left(\sum_{i=1}^n E(X_i - \mu)^2 - nE(\bar{X} - \mu)^2 \right) = \frac{1}{n} \left(n\sigma^2 - n\frac{\sigma^2}{n} \right) = \left(\frac{n-1}{n} \right) \sigma^2 \end{aligned}$$

- The sample variance, $s^2 \left(= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \right)$ is an unbiased estimator of the population variance σ^2 .

- Sketch of proof: $E[s^2] = E\left[\frac{n}{n-1} S_n^2\right] = \frac{n}{n-1} E[S_n^2] = \sigma^2$

- Unbiased estimators need not be unique.
 - Let $X_1, \dots, X_n$ be a random sample from a population with finite mean μ . Then the sample mean $\bar{X}$ and $\frac{1}{3}\bar{X} + \frac{2}{3}X_1$ are both unbiased estimators of μ .
 - Let $\hat{\theta}_1$ and $\hat{\theta}_2$ be two unbiased estimators of θ . Then $\hat{\theta}_3 = a\hat{\theta}_1 + (1 - a)\hat{\theta}_2, 0 \leq a \leq 1$ is an unbiased estimator of θ .

Mean Square Error of an Estimator

- The mean square error of the estimator $\hat{\theta}$, denoted by $MSE(\hat{\theta})$, is defined as

$$MSE(\hat{\theta}) = E(\hat{\theta} - \theta)^2.$$

- Theorem:** $MSE(\hat{\theta}) = Var(\hat{\theta}) + B^2(\hat{\theta})$.

- Sketch of proof:

$$\begin{aligned} MSE(\hat{\theta}) &= E(\hat{\theta} - \theta)^2 = E\left((\hat{\theta} - E(\hat{\theta})) + (E(\hat{\theta}) - \theta)\right)^2 \\ &= E\left(\hat{\theta} - E(\hat{\theta})\right)^2 + \left(E(\hat{\theta}) - \theta\right)^2 + 2\left(E(\hat{\theta}) - \theta\right)\left(E(\hat{\theta}) - E(\hat{\theta})\right) \\ &= Var(\hat{\theta}) + B^2(\hat{\theta}). \end{aligned}$$

- If $\hat{\theta}$ is an unbiased estimator of θ , then

$$B^2(\hat{\theta}) = 0 \text{ and } MSE(\hat{\theta}) = Var(\hat{\theta}).$$

- Minimum variance unbiased estimator (MVUE) of θ :
 - The unbiased estimator $\hat{\theta}$ that minimizes the mean square error is called the *MVUE* of θ .

Sufficient Estimators

- A statistic U is a sufficient statistic for a parameter θ if U contains all the information available in the data about the value of θ .
- If $U(\mathbf{X})$ is a sufficient statistic for a parameter θ , then any inference about θ should depend on the sample $\mathbf{X}$ only through the value $U(\mathbf{X})$.
 - That is, if $\mathbf{x}$ and $\mathbf{y}$ are two sample points such that $U(\mathbf{x}) = U(\mathbf{y})$, then the inference about θ should be the same whether $\mathbf{X} = \mathbf{x}$ or $\mathbf{X} = \mathbf{y}$ is observed.
- Example
 - the sample mean ($\bar{X}$) may contain all the relevant information about the parameter μ , and in that case $U(\mathbf{X}) = \bar{X}$ is called a sufficient statistic for μ .

Formal definition

- Let $X_1, \dots, X_n$ be a random sample from a probability distribution with unknown parameter θ .
 - Then, the statistic $U = g(X_1, \dots, X_n)$ is said to be **sufficient statistic** for θ if $f_{X_1, \dots, X_n}(x_1, \dots, x_n | U = u)$ does not depend on θ for any value of u .
 - An estimator of θ that is a function of a sufficient statistic for θ is said to be a **sufficient estimator** of θ .

Sufficient Estimators III

- Example: Let $X_1, \dots, X_n$ be i.i.d. Bernoulli random variables with parameter θ . Then $U = \sum_{i=1}^n X_i$ is sufficient for θ .

- Sketch of proof:

$$\begin{aligned} f(X_1, \dots, X_n; \theta) &= \theta^{\sum_{i=1}^n X_i} (1 - \theta)^{n - \sum_{i=1}^n X_i} \\ &= \theta^U (1 - \theta)^{n-U}. \end{aligned}$$

Since, $U \sim \text{Bin}(n, \theta)$,

$$f(U; \theta) = {}^n C_U \theta^U (1 - \theta)^{n-U}.$$

Thus,

$$f(x_1, \dots, x_n | U = u) = \frac{f(x_1, \dots, x_n, u)}{f_U(u)} = \begin{cases} \frac{1}{{}^n C_U} & ; \text{ if } u = \sum x_i \\ 0 & ; \text{ otherwise.} \end{cases}$$

Neyman–Fisher factorization theorem to *spot* a sufficient statistic.

- *Theorem:* -

Let U be a statistic based on the random sample $X_1, \dots, X_n$. Then, U is a sufficient statistic for θ if and only if the joint p.d.f or p.m.f., $f(x_1, \dots, x_n; \theta)$ can be factored into two non-negative functions, i.e.,

$$f(x_1, \dots, x_n; \theta) = g(u, \theta)h(x_1, \dots, x_n), \text{ for all } x_1, \dots, x_n,$$

where

- $g(u, \theta)$ is a function only of u and θ
- and $h(x_1, \dots, x_n)$ is a function of only $x_1, \dots, x_n$ and not of θ .

Sufficient Estimators V

- Sketch of proof: discrete case
 - Sufficient $\Rightarrow$ Factorization

$$\begin{aligned}f(x_1, \dots, x_n; \theta) &= P_\theta(X_1 = x_1, \dots, X_n = x_n, U = u) \\&= P_\theta(X_1 = x_1, \dots, X_n = x_n | U = u) P_\theta(U = u) \\&\stackrel{\text{uff.}}{=} h(x_1, \dots, x_n) g(u, \theta).\end{aligned}$$

- Factorization $\Rightarrow$ Sufficient

$$\begin{aligned}P_\theta(X_1 = x_1, \dots, X_n = x_n | U = u) &= \frac{P_\theta(X_1 = x_1, \dots, X_n = x_n, U = u)}{P_\theta(U = u)} \\&= \begin{cases} \frac{P_\theta(X_1 = x_1, \dots, X_n = x_n, U = u)}{P_\theta(U = u)} & \text{if } (x_1, \dots, x_n) \in A_u \\ 0 & \text{if } (x_1, \dots, x_n) \notin A_u, \end{cases}\end{aligned}$$

where A_u is the set of all $(x_1, \dots, x_n)$ such that U maps it into u .

Sufficient Estimators VI

- When $(x_1, x_2, \dots, x_n) \in A_u$,

$$\begin{aligned} P_\theta(X_1 = x_1, \dots, X_n = x_n | U = u) &= \frac{P_\theta(X_1 = x_1, \dots, X_n = x_n, U = u)}{P_\theta(U = u)} \\ &= \frac{P_\theta(X_1 = x_1, \dots, X_n = x_n)}{P_\theta(U = u)} \\ &= \frac{f(x_1, \dots, x_n, \theta)}{\sum_{(x_1, \dots, x_n) \in A_u} f(x_1, \dots, x_n, \theta)} \\ &\stackrel{\text{fact}}{=} \frac{g(u, \theta)h(x_1, \dots, x_n)}{\sum_{(x_1, \dots, x_n) \in A_u} g(u, \theta)h(x_1, \dots, x_n)} \\ &= \frac{h(x_1, \dots, x_n)}{\sum_{(x_1, \dots, x_n) \in A_u} h(x_1, \dots, x_n)} \perp \theta \end{aligned}$$

- When $(x_1, x_2, \dots, x_n) \notin A_u$,

$$\begin{aligned} P_\theta(X_1 = x_1, \dots, X_n = x_n | U = u) &= \frac{P_\theta(X_1 = x_1, \dots, X_n = x_n, U = u)}{P_\theta(U = u)} \\ &= 0 \perp \theta \end{aligned}$$

- Procedure to verify Sufficiency
 - 1 Obtain the joint pdf or pmf $f_{\theta}(x_1, \dots, x_n)$.
 - 2 If necessary, rewrite the joint pdf or pmf in terms of the given statistic and parameter so that one can use the factorization theorem.
 - 3 Define the functions g and h , in such a way that g is a function of the statistic and parameter only and h is a function of the observations only.
 - 4 If step 3 is possible, then the statistic is sufficient. Otherwise, it is not sufficient.

Sufficient Estimators VIII

- Example: Let $X_1, \dots, X_n$ denote a random sample from a geometric population with parameter p . Show that $\bar{X}$ is sufficient for p .

- Sketch of proof:

$$\begin{aligned} f(x_1, \dots, x_n) &= \prod_{i=1}^n p(1-p)^{x_i-1} \\ &= p^n (1-p)^{-n + \sum_{i=1}^n x_i} \\ &= p^n (1-p)^{-n + n\bar{x}} \\ &= g(\bar{x}, p) \underbrace{h(x_1, \dots, x_n)}_1 \end{aligned}$$

- *Joint Sufficiency:*

Two statistics U_1 and U_2 are said to be jointly sufficient for the parameters θ_1 and θ_2 if the conditional distribution of $X_1, \dots, X_n$ given U_1 and U_2 does not depend on θ_1 or θ_2 .

In general, the statistic $U = (U_1, \dots, U_n)$ is jointly sufficient for $\theta = (\theta_1, \dots, \theta_n)$ if the conditional distribution of $X_1, \dots, X_n$ given U is free of θ .

- Factorization criteria for Joint Sufficiency

Theorem: -

The two statistics U_1 and U_2 are jointly sufficient for θ_1 and θ_2 if and only if the likelihood function can be factored into two non-negative functions,

$$f(x_1, \dots, x_n; \theta_1, \theta_2) = g(u_1, u_2; \theta_1, \theta_2)h(x_1, \dots, x_n)$$

where $g(u_1, u_2; \theta_1, \theta_2)$ is only a function of $u_1, u_2; \theta_1$ and θ_2 , and $h(x_1, \dots, x_n)$ is free of θ_1 or θ_2

- Example: Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$.
 - ① If μ is unknown and $\sigma^2 = \sigma_0^2$ is known, then $U_1 = \bar{X}$ is a sufficient statistic for μ .

$$L = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma_0} e^{-\frac{1}{2}\left(\frac{x_i - \mu}{\sigma_0}\right)^2} = \underbrace{(2\pi\sigma_0^2)^{-n/2} e^{-\frac{1}{2\sigma_0^2} \left[\sum_{i=1}^n x_i^2 \right]}}_{h(x_1, \dots, x_n)} \times \underbrace{e^{-\frac{1}{2\sigma_0^2} [-2n\mu\bar{x} + n\mu^2]}}_{g(u_1, \theta_1)}$$

- 2 If $\mu = \mu_0$ is known and σ^2 is unknown, then $U_1 = \sum_{i=1}^n (X_i - \mu_0)^2$ is a sufficient statistic for σ^2 .

$$L = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x_i - \mu_0}{\sigma}\right)^2} = \underbrace{(2\pi)^{-n/2}}_{h(x_1, \dots, x_n)} \times \underbrace{\sigma^{-n} e^{-\frac{\sum_{i=1}^n (x_i - \mu_0)^2}{2\sigma^2}}}_{g(u_1, \theta_1)}$$

- 3 If μ and σ^2 are both unknown, then $U_1 = \sum_{i=1}^n X_i$ and $U_2 = \sum_{i=1}^n X_i^2$ are jointly sufficient for μ and σ^2 .

$$L = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x_i - \mu}{\sigma}\right)^2} = \underbrace{(2\pi)^{-n/2}}_{h(x_1, \dots, x_n)} \times \underbrace{\sigma^{-n} e^{-\frac{1}{2\sigma^2} \left[\sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2 \right]}}_{g(u_1, u_2, \theta_1, \theta_2)}$$

- *Theorem:* -

If U is a sufficient statistic for θ , then the maximum likelihood estimator of θ , if unique, is a function of U .

- Sketch of proof.

$$f(x_1, \dots, x_n; \theta) \stackrel{\text{suff}}{=} g(u, \theta)h(x_1, \dots, x_n).$$

Thus, the joint pdf/pmf depends on θ only through the statistic U . To maximize L we need to maximize $g(U, \theta)$.

Sufficient Estimators XV

- *Theorem:* -

Let $X_1, \dots, X_n$ be a random sample from a population with pdf or pmf of the exponential form

$$f(x_1, \dots, x_n; \theta) = \begin{cases} \exp[k(x)c(\theta) + S(x) + d(\theta)] & , x \in B \\ 0 & , x \notin B, \end{cases}$$

where B does not depend on the parameter θ . The statistic

$$U = \sum_{i=1}^n k(X_i) \text{ is sufficient for } \theta.$$

- Distributions in exponential form

- *Bernoulli*(p): $p^x(1-p)^{1-x} = \exp \left[x \ln\left(\frac{p}{1-p}\right) + \ln(1-p) \right]; x = 0, 1.$
- *Poisson*(λ): $\frac{\lambda^x e^{-\lambda}}{x!} = \exp [x \ln(\lambda) - \ln(x!) - \lambda]; x = 1, 2, \dots$
- *Normal*($\mu, 1$): $\frac{1}{\sqrt{2\pi}} e^{-(x-\mu)^2/2} = \exp \left[x\mu - \frac{x^2}{2} - \frac{\mu^2}{2} - \frac{\ln(2\pi)}{2} \right]; x \in \mathcal{R}.$

- Sketch of proof.

$$\begin{aligned} f(x_1, \dots, x_n; \theta) &= \exp \left[c(\theta) \sum_{i=1}^n k(x_i) + \sum_{i=1}^n S(x_i) + nd(\theta) \right] \\ &= \underbrace{\exp \left[c(\theta) \sum_{i=1}^n k(x_i) + nd(\theta) \right]}_{g(u, \theta)} \underbrace{\exp \left[\sum_{i=1}^n S(x_i) \right]}_{h(x_1, \dots, x_n)}. \end{aligned}$$

Some Observations on Sufficiency

- 1 All the statistics need not be sufficient, in addition
- 2 Any function of a sufficient statistic needs not to be sufficient, however
- 3 Any one-to-one function of a sufficient statistic is also sufficient statistic

Sufficient Estimators XVIII

- *RAO-BLACKWELL Theorem:* -

Let $X_1, \dots, X_n$ be a random sample with joint pmf or pdf $f(x_1, \dots, x_n; \theta)$ and let $U = (U_1, \dots, U_n)$ be jointly sufficient for $\theta = (\theta_1, \dots, \theta_n)$. If T is any unbiased estimator of $k(\theta)$, and if $T^* = E(T|U)$, then:

- T^* is an unbiased estimator of $k(\theta)$.

Sketch of proof: - $ET^* = E(E(T|U)) = E(T) = k(\theta)$.

Hence, T^* is an unbiased estimator of $k(\theta)$.

- T^* is a function of U , and does not depend on θ .

Sketch of proof: - Because U is sufficient for θ , the conditional distribution of any statistic (hence, for T), given U , does not depend on θ .

- $Var(T^*) \leq Var(T)$ for every θ , and $Var(T^*) < Var(T)$ for some θ unless $T^* = T$ with probability 1.

Sketch of proof: - $Var(T) = E(Var(T|U)) + Var(E(T|U)) = E(Var(T|U)) + Var(T^*)$.

Because $Var(T|U) \geq 0$ for all u , it follows that $E(Var(T|U)) \geq 0$. Hence, $Var(T^*) \leq Var(T)$.

Also $Var(T^*) = Var(T)$ iff $Var(T|U) = 0$ or T is a function of U , in which case $T^* = E(T|U) = T$.

More Observations

- If one is searching for an unbiased estimator with minimal variance, it has to be restricted to functions of a sufficient statistics.
- If $k(\theta) = \theta$, and T is an unbiased estimator of θ , then $T^* = E(T|U)$ will typically give the MVUE of θ .

- Minimal sufficient statistic:

A sufficient statistic $T(X)$ is called a minimal sufficient statistic if for any other sufficient statistic $T'(X)$,

$$T(X) = g(T'(X)),$$

i.e., $T(X)$ is a function of $T'(X)$.

- Intuitively, a minimal sufficient statistic most efficiently captures all possible information about the parameter θ .

- Let $f(\mathbf{x}|\theta)$ be the pmf/pdf of a sample $\mathbf{X}$. Suppose there exists a function $T(\mathbf{x})$ such that, for every two sample points $\mathbf{x}$ and $\mathbf{y}$, the ratio

$$\frac{f(\mathbf{x}|\theta)}{f(\mathbf{y}|\theta)}$$

is constant as a function of θ if and only if

$$T(\mathbf{x}) = T(\mathbf{y}).$$

Then $T(\mathbf{X})$ is a minimal sufficient statistic for θ .

Sufficient Estimators XXII

- Lehmann and Scheffe method to find a minimal sufficient statistic
 - Let $X_1, \dots, X_n$ be a random sample with pdf or pmf $f(x)$ that depends on a parameter θ .
 - Let $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ be two different sets of values of $(X_1, \dots, X_n)$.
 - Let $\frac{L(\theta; x_1, \dots, x_n)}{L(\theta; y_1, \dots, y_n)}$ be the ratio of the likelihoods evaluated at these two points.
 - Suppose it is possible to find a function $g(x_1, \dots, x_n)$ such that this ratio will be free of the unknown parameter θ if and only if $g(x_1, \dots, x_n) = g(y_1, \dots, y_n)$,
 - in other words

$$\frac{L(\theta; x_1, \dots, x_n)}{L(\theta; y_1, \dots, y_n)} \text{ is independent of } \theta \Leftrightarrow g(x_1, \dots, x_n) = g(y_1, \dots, y_n).$$

- If such a function g can be found, then $g(X_1, \dots, X_n)$ is a minimal sufficient statistic for θ .

- Example 1: -

Let $(X_1, \dots, X_n)$ be a random sample from the *Bernoulli*(p), where p is unknown. Find a minimal sufficient statistic for p .

- Solution: - The ratio of the likelihoods is

$$\frac{L(x_1, \dots, x_n)}{L(y_1, \dots, y_n)} = \frac{p^{x_1} (1-p)^{1-x_1} \dots p^{x_n} (1-p)^{1-x_n}}{p^{y_1} (1-p)^{1-y_1} \dots p^{y_n} (1-p)^{1-y_n}} = \frac{p^{\sum x_i} (1-p)^{n-\sum x_i}}{p^{\sum y_i} (1-p)^{n-\sum y_i}} = \left(\frac{p}{1-p} \right)^{\sum x_i - \sum y_i}$$

This ratio is to be independent of p , if and only if

$$g(x_1, \dots, x_n) = \sum_{i=1}^n x_i = \sum_{i=1}^n y_i = g(y_1, \dots, y_n).$$

Therefore, $g(X_1, \dots, X_n) = \sum_{i=1}^n X_i$ is a minimal sufficient statistic for p .

- Example 2: -

Let $(X_1, \dots, X_n)$ be a random sample from the $Normal(\mu, \sigma^2)$, where μ and σ^2 are unknown. Show that $(\bar{X}, S^2)$ is a minimal sufficient statistic for (μ, σ^2) .

- Solution: - The ratio of the likelihoods is

$$\begin{aligned} \frac{L(x_1, \dots, x_n)}{L(y_1, \dots, y_n)} &= \frac{(2\pi\sigma^2)^{-n/2} \exp(-[n(\bar{x} - \mu)^2 + (n-1)S_x^2]/(2\sigma^2))}{(2\pi\sigma^2)^{-n/2} \exp(-[n(\bar{y} - \mu)^2 + (n-1)S_y^2]/(2\sigma^2))} \\ &= \exp\left(\frac{-n[\bar{x}^2 - \bar{y}^2] + 2n\mu[\bar{x} - \bar{y}] - (n-1)[S_x^2 - S_y^2]}{2\sigma^2}\right) \end{aligned}$$

This ratio is to be independent of (μ, σ^2) , if and only if

$$\bar{x} = \bar{y} \text{ and } S_x^2 = S_y^2$$

Therefore, $(\bar{X}, S^2)$ is a minimal sufficient statistic for (μ, σ^2) .

- Note that

- The dimension of a minimal sufficient statistic may not match the dimension of the parameter.

- Example 3: -

Let $(X_1, \dots, X_n)$ be a random sample from the $Uniform(\theta, \theta + 1)$, where $-\infty < \theta < \infty$ is unknown. Find a minimal sufficient statistic for θ .

- Solution: - The joint pdf of $\mathbf{X}$ is

$$f(x_1, \dots, x_n; \theta) = \begin{cases} 1 & , \theta < x_i < \theta + 1, i = 1, \dots, n. \\ 0 & , \text{otherwise.} \end{cases}$$

which can be rewritten as

$$f(x_1, \dots, x_n; \theta) = \begin{cases} 1 & , \max_i x_i - 1 < \theta < \min_i x_i. \\ 0 & , \text{otherwise.} \end{cases}$$

Thus, for two sample points $\mathbf{x}$ and $\mathbf{y}$, the numerator and denominator of the ratio $f(\mathbf{x}|\theta)/f(\mathbf{y}|\theta)$ will be positive for the same values of θ if and only if $\min_i X_i = \min_i y_i$ and $\max_i X_i = \max_i y_i$. And, if the minima and maxima are equal, then the ratio is constant and, in fact, equals 1.

Thus, letting $X_{(1)} = \min_i X_i$ and $X_{(n)} = \max_i X_i$, we have that $T(\mathbf{X}) = (X_{(1)}, X_{(n)})$ is a minimal sufficient statistic.

- A minimal sufficient statistic is not unique.
- Any one-to-one function of a minimal sufficient statistic is also a minimal sufficient statistic.
 - Thus $(R = X_{(n)} - X_{(1)}, M = (X_{(n)} + X_{(1)}) / 2)$, is also a minimal sufficient statistic in Example 3

- Ancillary Statistic: - A statistic $T(\mathbf{X})$ whose distribution does not depend on the parameter θ is called an ancillary statistic for θ .

- Example: -

Let $(X_1, \dots, X_n)$ be a random sample from the $Uniform(\theta, \theta + 1)$, where $-\infty < \theta < \infty$ is unknown. One can show $R = x_{(n)} - x_{(1)}$ is an ancillary statistic for θ .

Sufficient Estimators XXIX

- Solution: - The cdf of X_i is

$$F(x|\theta) = \begin{cases} 0 & , x \leq \theta, \\ x - \theta & , \theta < x < \theta + 1, \\ 1 & , \theta + 1 \leq x. \end{cases}$$

Thus the joint pdf of $(x_{(1)}, x_{(n)})$ is

$$g(x_{(1)}, x_{(n)}|\theta) = \begin{cases} n(n-1)(x_{(n)} - x_{(1)})^{n-2} & , \theta < x_{(1)} < x_{(n)} < \theta + 1. \\ 0 & , \text{otherwise.} \end{cases}$$

Making the transformation $R = X_{(n)} - X_{(1)}$ and $M = (X_{(n)} + X_{(1)})/2$, which has the inverse transformation $X_{(1)} = (2M - R)/2$ and $X_{(n)} = (2M + R)/2$, with Jacobian equals 1. Thus, the joint pdf of (R, M) is

$$h(r, m|\theta) = \begin{cases} n(n-1)r^{n-2} & , 0 < r < 1, \theta + r/2 < m < \theta + 1 - r/2. \\ 0 & , \text{otherwise.} \end{cases}$$

Thus, the pdf for R is

$$h(r|\theta) = \int_{\theta+r/2}^{\theta+1-r/2} n(n-1)r^{n-2} dm = n(n-1)r^{n-2}(1-r); 0 < r < 1.$$

Thus, the distribution of R does not depend on θ , and R is ancillary.

- Note that

- Since the distribution of an ancillary statistic does not depend on θ , it might be suspected that a minimal sufficient statistic is unrelated to (or mathematically speaking, functionally independent of) an ancillary statistic.
- However, this is not necessarily the case. As we have seen in the last two examples that R is ancillary as well as part of minimal sufficient statistic.
- Hence, the ancillary statistic and the minimal sufficient statistic are not independent.
- For many important situations, however, the intuition that a minimal sufficient statistic is independent of any ancillary statistic is correct.
 - A minimal sufficient statistic is independent of any ancillary statistic if it is a complete statistic as well.

- Complete Statistic: -

Let $f(t|\theta)$ be a family of pdfs/pmfs for a statistic $T(\mathbf{X})$. The family of probability distributions is called complete if $E_{\theta}g(T) = 0$ for all θ implies $P_{\theta}(g(T) = 0) = 1$ for all θ .

Equivalently, $T(\mathbf{X})$ is called a complete statistic.

- Note that

- Completeness is a property of a family of probability distributions, not of a particular distribution.
- If $X \sim N(0, 1)$, then defining $g(x) = x$, we have $Eg(X) = 0$. But the function $g(x) = x$ satisfies $P(g(X) = 0) = 0$, not 1.
- However, this is a particular distribution, not a family of distributions.
- If $X \sim N(\theta, 1)$, $-\infty < \theta < \infty$, we can see that no function of X , except one that is 0 with probability 1 for all θ , satisfies $E_{\theta}g(X) = 0$ for all θ .
 - Thus, the family of $N(\theta, 1)$ distributions, $-\infty < \theta < \infty$, is complete.

Sufficient Estimators XXXIII

- Binomial complete sufficient statistic: -
Suppose $T \sim \text{Binomial}(n, p)$, with $0 < p < 1$. Let g be a function such that $E_p g(T) = 0$. Then,

$$\begin{aligned} 0 &= E_p g(T) = \sum_{t=0}^n g(t) \binom{n}{t} p^t (1-p)^{n-t} \\ &= (1-p)^n \sum_{t=0}^n g(t) \binom{n}{t} \left(\frac{p}{1-p} \right)^t \end{aligned}$$

for all $p, 0 < p < 1$. Thus it must be that

$$0 = \sum_{t=0}^n g(t) \binom{n}{t} \left(\frac{p}{1-p} \right)^t \Leftrightarrow g(t) = 0, \text{ for all } t = 0, \dots, n.$$

Since, T takes values $0, 1, \dots, n$ with probability 1, this yields that $P_p(g(T) = 0) = 1$, for all p . Hence, T is a complete statistic.

Sufficient Estimators XXXIV

- Uniform complete sufficient statistic: -

Let $X_1, \dots, X_n$ be iid $Uniform(0, \theta)$ observations, $0 < \theta < \infty$. We have shown that $T(X) = \max_i X_i$ is a sufficient statistic and pdf of $T(X)$, is

$$f(t|\theta) = \begin{cases} n(t/\theta)^{n-1} 1/\theta & , x \leq \theta, \\ 0 & , otherwise. \end{cases}$$

Suppose $g(t)$ is a function satisfying $E_\theta g(T) = 0$ for all θ . Since $E_\theta g(T)$ is constant as a function of θ , its derivative with respect to θ is 0. Thus we have that

$$\begin{aligned} 0 &= \frac{d}{d\theta} E_\theta g(T) = \frac{d}{d\theta} \int_0^\theta g(t) n t^{n-1} \theta^{-n} dt \\ &= n g(\theta) \theta^{n-1} \theta^{-n} + \int_0^\theta \frac{\delta}{\delta \theta} g(t) n t^{n-1} \theta^{-n} dt = n g(\theta) \theta^{-1}. \end{aligned}$$

Since, $n g(\theta) \theta^{-1} = 0$ and $n \theta^{-1} \neq 0$, it must be that $g(\theta) = 0$, for every $\theta > 0$. Hence, T is a complete statistic.

- Basu's Theorem

If $T(X)$ is a complete and minimal sufficient statistic, then $T(X)$ is independent of every ancillary statistic.

- Note that

- To show that a statistic is complete, which is sometimes a rather difficult analysis problem.
- Fortunately, most problems we are concerned with are covered by the following theorem.

- Complete statistics in the exponential family: -
Let $X_1, \dots, X_n$ be iid observations from an exponential family with pdf or pmf of the form

$$f(\mathbf{x}|\underline{\theta}) = h(\mathbf{x})c(\underline{\theta}) \exp \left(\sum_{j=1}^k w(\theta_j)t_j(\mathbf{x}) \right),$$

where $\underline{\theta} = (\theta_1, \dots, \theta_k)$ then the statistic

$$T(\mathbf{X}) = \left(\sum_{i=1}^n t_1(X_i), \sum_{i=1}^n t_2(X_i), \dots, \sum_{i=1}^n t_k(X_i) \right)$$

is complete as long as the parameter space Θ contains an open set in $\mathcal{R}^k$.

- Application of Basu's Theorem

- Independence of $\bar{X}$ and S , the sample mean and variance, when sampling from a $N(\mu, \sigma^2)$ population.
 - For known σ^2 , and unknown $\mu \in (-\infty, \infty)$, $\bar{X}$ is sufficient statistic for μ .
 - In addition, the family of $N(\mu, \sigma^2)$ distributions, where $-\infty < \mu < \infty$, and with known $\frac{\sigma^2}{n}$, is a complete family. Since this is the distribution of $\bar{X}$, $\bar{X}$ is a complete statistic.
 - The S is ancillary statistic for μ , since $(n-1)\frac{S^2}{\sigma^2} \sim \chi_{n-1}^2$.

- Note

- If a minimal sufficient statistic exists, then any complete statistic is also a minimal sufficient statistic.
- So even though the word “minimal” is redundant in the statement of Basu’s Theorem, it was stated in this way as a reminder that the statistic $T(X)$ in the theorem is a minimal sufficient statistic.

Consistent Estimators

- A statistic is a consistent estimator if its value becomes closer to the value of the true parameter which is being estimated, as the sample size becomes larger.

Consistent Estimators II

Formal definition

- The estimator $\hat{\theta}_n$ is said to be a consistent estimator of θ if, for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P \left[|\hat{\theta}_n - \theta| \leq \epsilon \right] = 1$$

or equivalently,

$$\lim_{n \rightarrow \infty} P \left[|\hat{\theta}_n - \theta| > \epsilon \right] = 0,$$

i.e. $\hat{\theta}_n$ converges in probability to θ .

Consistent Estimators III

- Example: - Let $X_1, \dots, X_n$ be a random sample with true mean μ and finite variance, σ^2 . Then, the sample mean $\bar{X}$ is a consistent estimator of the population mean μ .

- Sketch of proof: -
Note that, for any positive r.v. X (Markov Inequality)

$$E[X] = \int_0^\epsilon x f_X(x) dx + \int_\epsilon^\infty x f_X(x) dx \geq \int_0^\epsilon x f_X(x) dx + \epsilon \int_\epsilon^\infty f_X(x) dx \geq \epsilon P(X \geq \epsilon).$$

Hence, (Chebyshev's Inequality)

$$P[|\bar{X} - \mu| \geq \epsilon] = P[(\bar{X} - E[\bar{X}])^2 \geq \epsilon^2] \leq \frac{E(\bar{X} - E[\bar{X}])^2}{\epsilon^2} = \frac{\text{Var}(\bar{X})}{\epsilon^2},$$

Thus,

$$\lim_{n \rightarrow \infty} P[|\bar{X} - \mu| \geq \epsilon] \leq \lim_{n \rightarrow \infty} \frac{\text{Var}(\bar{X})}{\epsilon^2} = \lim_{n \rightarrow \infty} \frac{\sigma^2}{n\epsilon^2} = 0$$

Test for Consistency (Sufficient conditions)

- *Theorem:* - An unbiased estimator $\hat{\theta}_n$ of θ is a consistent estimator for θ if

$$\lim_{n \rightarrow \infty} \text{Var}(\hat{\theta}_n) = 0.$$

- Sketch of proof: - $\lim_{n \rightarrow \infty} P[|\hat{\theta}_n - \theta| \geq \epsilon] \stackrel{U.E.}{=} \lim_{n \rightarrow \infty} P[|\hat{\theta}_n - E(\hat{\theta}_n)| \geq \epsilon] \stackrel{C.I.}{\leq} \lim_{n \rightarrow \infty} \frac{\text{Var}(\hat{\theta}_n)}{\epsilon^2} = 0$
- An unbiased estimator $\hat{\theta}_n$ is consistent if $\text{Var}(\hat{\theta}_n) \rightarrow 0$ as $n \rightarrow \infty$.

(Contd.)

- **Theorem:** - An estimator $\hat{\theta}_n$, with finite variance, is a consistent estimator for θ if

$$\lim_{n \rightarrow \infty} E(\hat{\theta}_n - \theta)^2 = 0.$$

- Sketch of proof: - $\lim_{n \rightarrow \infty} P[|\hat{\theta}_n - \theta| \geq \epsilon] \stackrel{C.I.}{\leq} \lim_{n \rightarrow \infty} \frac{E(\hat{\theta}_n - \theta)^2}{\epsilon^2} = 0$
- A biased estimator $\hat{\theta}_n$ is consistent if both $\text{Var}(\hat{\theta}_n) \rightarrow 0$ and $B(\hat{\theta}_n) \rightarrow 0$ as $n \rightarrow \infty$.

Consistent Estimators VI

- Example: -

Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$ population.

- Then the sample variance S^2 is a consistent estimator for σ^2 .

- Sketch of proof: -

$$(n-1) \frac{S^2}{\sigma^2} \sim \chi^2(n-1) \Rightarrow \text{Var} \left[(n-1) \frac{S^2}{\sigma^2} \right] = 2(n-1) \Rightarrow \text{Var}(S^2) = \frac{2\sigma^4}{n-1}.$$

S^2 is u.e. of σ^2 and $\lim_{n \rightarrow \infty} \text{Var}(S^2) = 0$.

- The maximum likelihood estimators $\bar{X}$ and S_n^2 for μ and σ^2 , respectively, are consistent estimators for μ and σ^2 .

- Sketch of proof: -

$\bar{X}$ is u.e. of μ and $\lim_{n \rightarrow \infty} \text{Var}(\bar{X}) = 0$, thus $\bar{X}$ is consistent estimator of μ .

$$\text{Var}(S_n^2) = \text{Var} \left(\frac{n-1}{n} S^2 \right) = \frac{2(n-1)\sigma^4}{n^2}, \text{ thus } \lim_{n \rightarrow \infty} \text{Var}(S_n^2) = 0 \text{ and}$$

$$B(S_n^2) = \frac{n-1}{n} \sigma^2 - \sigma^2 = -\frac{1}{n} \sigma^2, \text{ thus } \lim_{n \rightarrow \infty} B(S_n^2) = 0$$

Efficiency

- It is a relative comparison between variances of two unbiased/biased estimators

Formal Definitions

- If $\hat{\theta}_1$ and $\hat{\theta}_2$ are two unbiased estimators for θ , the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is the ratio

$$e(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}.$$

- If $\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2)$, or equivalently, $e(\hat{\theta}_1, \hat{\theta}_2) > 1$, then, $\hat{\theta}_1$ is relatively more efficient than $\hat{\theta}_2$.

(Contd.)

- If $\hat{\theta}_1$ and $\hat{\theta}_2$ are two biased estimators for θ , the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is the ratio

$$e(\hat{\theta}_1, \hat{\theta}_2) = \frac{E(\hat{\theta}_2 - \theta)^2}{E(\hat{\theta}_1 - \theta)^2} = \frac{MSE(\hat{\theta}_2)}{MSE(\hat{\theta}_1)}.$$

- If $MSE(\hat{\theta}_1) < MSE(\hat{\theta}_2)$, or equivalently, $e(\hat{\theta}_1, \hat{\theta}_2) > 1$, then, $\hat{\theta}_1$ is relatively more efficient than $\hat{\theta}_2$.

- Example:-

Let $X_1, \dots, X_n$, $n \geq 2$ be a random sample from a normal population with a true mean μ and variance σ^2 . Consider the following two estimators of σ^2 : $\theta_1 = S^2$, and $\theta_2 = S_n^2$. Find $e(\theta_1, \theta_2)$.

- Sketch of proof: -

$$MSE(\hat{\theta}_1) = 0 + \frac{2\sigma^4}{n-1} = \frac{2\sigma^4}{n-1}$$

$$MSE(\hat{\theta}_2) = \left(-\frac{\sigma^2}{n}\right)^2 + \frac{2(n-1)\sigma^4}{n^2} = \frac{(2n-1)\sigma^4}{n^2}$$

$$e(\hat{\theta}_1, \hat{\theta}_2) = \frac{(n-1)(2n-1)}{2n^2} < 1, \text{ for } n \geq 2.$$

Hence, S_n^2 is relatively more efficient than S^2 .

Uniformly Minimum Variance Unbiased Estimator

- An unbiased estimator $\hat{\theta}_0$, is said to be a uniformly minimum variance unbiased estimator (UMVUE) for the parameter θ if, for any other unbiased estimator $\hat{\theta}$,

$$\text{Var}(\hat{\theta}_0) \leq \text{Var}(\hat{\theta}),$$

for all possible values of θ .

Cramer–Rao Inequality:

(Lower bound for the variance of any estimator)

- *Theorem:* - Let $X_1, \dots, X_n$ be a sample with pdf $f(\mathbf{x}|\theta)$, and let $W(\mathbf{X}) = W(X_1, \dots, X_n)$ be any estimator satisfying

$$\frac{d}{d\theta} E_{\theta}[W(\mathbf{X})] = \int_{\mathcal{X}} \frac{\delta}{\delta\theta} [W(\mathbf{x})f(\mathbf{x}|\theta)] d\mathbf{x}$$

and

$$\text{Var}_{\theta}[W(\mathbf{X})] < \infty,$$

then

$$\text{Var}_{\theta}[W(\mathbf{X})] \geq \frac{\left(\frac{d}{d\theta} E_{\theta}[W(\mathbf{X})]\right)^2}{E_{\theta}\left[\left(\frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta)\right)^2\right]}.$$

Efficiency VII

- Sketch of proof: -
Note that

$$\begin{aligned}\frac{d}{d\theta} E_{\theta}[W(\mathbf{X})] &= \int_{\mathcal{X}} \frac{\delta}{\delta\theta} [W(\mathbf{x})f(\mathbf{x}|\theta)] d\mathbf{x} \\&= \int_{\mathcal{X}} W(\mathbf{x}) \left[\frac{\delta}{\delta\theta} f(\mathbf{x}|\theta) \right] d\mathbf{x} \\&= \int_{\mathcal{X}} W(\mathbf{x}) \left[\frac{\delta f(\mathbf{x}|\theta)}{\delta\theta} \frac{1}{f(\mathbf{x}|\theta)} \right] f(\mathbf{x}|\theta) d\mathbf{x} \\&= E_{\theta} \left[W(\mathbf{X}) \frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta) \right] \\&= \text{Cov}_{\theta} \left(W(\mathbf{X}), \frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta) \right),\end{aligned}$$

$$\text{since } E_{\theta} \left[\frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta) \right] = \int_{\mathcal{X}} \left[\frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta) \right] f(\mathbf{x}|\theta) d\mathbf{x} = \int_{\mathcal{X}} \left[\frac{\delta}{\delta\theta} f(\mathbf{x}|\theta) \right] d\mathbf{x} = \frac{d}{d\theta} \left[\int_{\mathcal{X}} f(\mathbf{x}|\theta) d\mathbf{x} \right] = 0.$$

Also

$$\text{Var}_{\theta} \left(\frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta) \right) = E_{\theta} \left[\left(\frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta) \right)^2 \right].$$

Now,

$$\frac{\left[\text{Cov}_{\theta} \left(W(\mathbf{X}), \frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta) \right) \right]^2}{\text{Var}_{\theta} [W(\mathbf{X})] \text{Var}_{\theta} \left[\frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta) \right]} \leq 1 \Rightarrow \frac{\left[\frac{d}{d\theta} E_{\theta}[W(\mathbf{X})] \right]^2}{E_{\theta} \left[\left(\frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta) \right)^2 \right]} \leq \text{Var}_{\theta} [W(\mathbf{X})]$$

- *Corollary:* -

Let $X_1, \dots, X_n$ be an iid random sample from a population with pdf or pmf $f_\theta(x)$ that depends on a parameter θ . If $\hat{\theta} = W(\mathbf{x})$ is an unbiased estimator of $\psi(\theta)$, then

$$\text{Var}(\hat{\theta}) \geq \frac{\left[\frac{d}{d\theta}\psi(\theta)\right]^2}{nE\left[\left(\frac{\delta}{\delta\theta}\ln f_\theta(x)\right)^2\right]}.$$

- Sketch of proof: - Note that

$$\begin{aligned} E \left[\left(\frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta) \right)^2 \right] &= E \left[\left(\frac{\delta}{\delta\theta} \ln \prod_{i=1}^n f_{\theta}(x_i) \right)^2 \right] = E \left[\left(\sum_{i=1}^n \frac{\delta}{\delta\theta} \ln f_{\theta}(x_i) \right)^2 \right] \\ &= \sum_{i=1}^n E \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x_i) \right)^2 \right] + \sum_{i \neq j} E \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x_i) \right) \left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x_j) \right) \right] \\ &= nE \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right)^2 \right] + \sum_{i \neq j} E \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x_i) \right) \right] E \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x_j) \right) \right] \\ &= nE \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right)^2 \right] \end{aligned}$$

and

$$\left[\frac{d}{d\theta} E_{\theta}[W(\mathbf{x})] \right]^2 = \left[\frac{d}{d\theta} E_{\theta}[\hat{\theta}] \right]^2 = \left[\frac{d}{d\theta} \psi(\theta) \right]^2.$$

Hence,

$$\text{Var}(\hat{\theta}) \geq \frac{\left[\frac{d}{d\theta} E_{\theta}[\hat{\theta}] \right]^2}{E_{\theta} \left[\left(\frac{\delta}{\delta\theta} \ln f(\mathbf{x}|\theta) \right)^2 \right]} = \frac{\left[\frac{d}{d\theta} \psi(\theta) \right]^2}{nE \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right)^2 \right]}.$$

- *Corollary:* -

Let $X_1, \dots, X_n$ be an iid random sample from a population with pdf or pmf $f_\theta(x)$ that depends on a parameter θ . If $\hat{\theta}$ is an unbiased estimator of θ , then

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nE \left[\left(\frac{\delta}{\delta\theta} \ln f_\theta(x) \right)^2 \right]}.$$

Efficient Estimator

- If $\hat{\theta}$ is an unbiased estimator of θ and if

$$Var(\hat{\theta}) = \frac{1}{nE \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right)^2 \right]}$$

then $\hat{\theta}$ is a uniformly minimum variance unbiased estimator (UMVUE) of θ .

Sometimes $\hat{\theta}$ is also referred to as an efficient estimator.

- **Result:-** If the function $f(\cdot)$ is sufficiently smooth, specifically if $\frac{d}{d\theta} E_{\theta} \left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right) = \int \frac{\delta}{\delta\theta} \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right) f_{\theta}(x) \right] dx$, then

$$E_{\theta} \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right)^2 \right] = -E_{\theta} \left(\frac{\delta^2}{\delta\theta^2} \ln f_{\theta}(x) \right) = \text{Var} \left[\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right]$$

and for an unbiased estimator $\hat{\theta}$ for θ the Cramer-Rao inequality can be rewritten as

$$\text{Var}(\hat{\theta}) \geq \frac{1}{-nE \left(\frac{\delta^2}{\delta\theta^2} \ln f_{\theta}(x) \right)} = \frac{1}{n\text{Var} \left[\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right]}.$$

Efficiency XIII

- Sketch of proof: -

$$\begin{aligned} E_{\theta} \left(\frac{\delta^2}{\delta\theta^2} \ln f_{\theta}(x) \right) &= E_{\theta} \left[\frac{\delta}{\delta\theta} \left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right) \right] = E_{\theta} \left[\frac{\delta}{\delta\theta} \left(\frac{\frac{\delta}{\delta\theta} f_{\theta}(x)}{f_{\theta}(x)} \right) \right] \\ &= E_{\theta} \left[\left(\frac{\frac{\delta^2}{\delta\theta^2} f_{\theta}(x)}{f_{\theta}(x)} \right) - \left(\frac{\frac{\delta}{\delta\theta} f_{\theta}(x)}{f_{\theta}(x)} \right)^2 \right] \\ &= \int \frac{\delta^2}{\delta\theta^2} f_{\theta}(x) dx - E_{\theta} \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right)^2 \right] \\ &\stackrel{\text{assumption}}{=} \frac{\delta}{\delta\theta} \int \frac{\delta}{\delta\theta} f_{\theta}(x) dx - E_{\theta} \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right)^2 \right] \\ &= \frac{d}{d\theta} E_{\theta} \left[\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right] - E_{\theta} \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right)^2 \right] \\ &= -E_{\theta} \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right)^2 \right] \end{aligned}$$

Hence,

$$\text{Var}(\hat{\theta}) \geq \frac{1}{n E_{\theta} \left[\left(\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right)^2 \right]} = \frac{1}{-n E_{\theta} \left(\frac{\delta^2}{\delta\theta^2} \ln f_{\theta}(x) \right)} = \frac{1}{n \text{Var}_{\theta} \left[\frac{\delta}{\delta\theta} \ln f_{\theta}(x) \right]}.$$

- Example: - Let $X_1, \dots, X_n$ be a random sample from an $N(\mu, \sigma^2)$ population. Then $\bar{X}$ is an efficient estimator for μ .

- Sketch of proof: -

$$l(x, \mu) = \ln f(x, \mu) = c - \frac{(x - \mu)^2}{2\sigma^2}.$$

Thus,

$$\frac{\partial}{\partial \mu} l(x, \mu) = \frac{x - \mu}{\sigma^2} \text{ and } \frac{\partial^2}{\partial \mu^2} l(x, \mu) = -\frac{1}{\sigma^2}.$$

Hence,

$$\frac{1}{nE \left[\left(\frac{\partial}{\partial \theta} \ln f_{\theta}(x) \right)^2 \right]} = \frac{1}{nE \left[\left(\frac{\partial}{\partial \mu} l(x, \mu) \right)^2 \right]} = \frac{1}{-nE \left[\frac{\partial^2}{\partial \mu^2} l(x, \mu) \right]} = \frac{\sigma^2}{n} = \text{Var}(\bar{X})$$

Note: -

- For a given problem UMVUE may not exist.
- Even when an UMVUE exists, it is not necessary that it have a variance equal to the Cramer–Rao lower bound.
- The term $I(\theta) = E \left[\left(\frac{\delta}{\delta \theta} \ln f_{\theta}(x) \right)^2 \right]$ is called the Fisher information.
- It can be shown that the Fisher information in a sample of size n , denoted by $I_n(\theta)$, is n times the Fisher information in one observation. That is,

$$I_n(\theta) = nI(\theta).$$

Chapter 8b: Interval Estimation

Sudipta Das

- We estimate the parameter θ by an **interval** of values

- Let

$$L = T_1(X_1, X_2, \dots, X_n) \text{ and } U = T_2(X_1, X_2, \dots, X_n)$$

be two statistics such that

$$P[L \leq \theta \leq U] = 1 - \alpha,$$

where α is a pre-assigned small quantity. Usually, we take $\alpha = 0.05$ or 0.01 etc.

- The number $1 - \alpha$ is called the confidence coefficient
 - and the limits U and L , are called the upper and lower confidence limits, respectively.

Interval Estimation II

- The interval (L, U) is referred to as a $(1 - \alpha)100\%$ confidence interval (CI) of the parameter θ .
- For example, if $\alpha = 0.05$, then $P[L \leq \theta \leq U] = 0.95$.
 - Hence, the observed values of $[L, U] = [l, u]$, (say), is called a 95% confidence interval of θ .

Interval Estimation III

- An interval (L, U) should have two properties:
 - $P(L < \theta < U)$ is high, that is, the true parameter θ is in (L, U) with high probability, and
 - the length of the interval (L, U) should be relatively narrow on the average.
- In summary,
 - Interval estimation goes a step beyond point estimation by providing, in addition to the estimating interval (L, U) , a measure of one's confidence in the accuracy of the estimate.

Interval Estimation I

- Preliminary Notations

Let $(x_1, \dots, x_n)$ be the random sample drawn from the population of interest.

① n : Sample size

② $\bar{x}$: Sample mean (Calculated)

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

③ s^2 : Sample variance (Calculated)

$$s^2 = \frac{1}{n-1} [(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2]$$

④ $\hat{p}$: Sample proportion (Calculated)

$$\hat{p} = \frac{\text{Total No. of sample units having the specific characteristic}}{n}$$

Interval Estimation II

Estimating population mean (μ)

- Pivotal quantity (Point Estimate): Sample mean ($\bar{x}$)
- Intervals:
 - $\left[\bar{x} - z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right]$, where $z_{1-\frac{\alpha}{2}}$ is the $(1-\alpha/2)^{th}$ quantile from the standard normal distribution
 - Under the assumption: Population is Normal and variance (σ^2) is known
 - $\left[\bar{x} - t_{1-\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}, \bar{x} + t_{1-\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}} \right]$, where $t_{1-\frac{\alpha}{2}, n-1}$ is the $(1-\alpha/2)^{th}$ quantile from a t -distribution with $n-1$ d.f.
 - Under the assumption: Population is Normal and variance is unknown
 - $\left[\bar{x} - z_{1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}}, \bar{x} + z_{1-\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \right]$, where $z_{1-\frac{\alpha}{2}}$ is the $(1-\alpha/2)^{th}$ quantile from the standard normal distribution
 - Under the assumption: Sample size (n) is large

Interval Estimation III

- Example 6.2.3; (Page 302)
- Example 6.3.1; (Page 311)

Interval Estimation IV

Estimating population variance (σ^2)

- Pivotal quantity (Point Estimate): Sample variance (s^2)
- Intervals:
 - $\left[\frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}}, \frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2}, n-1}} \right]$, where $\chi^2_{1-\frac{\alpha}{2}, n-1}$ and $\chi^2_{\frac{\alpha}{2}, n-1}$ are the $(1-\alpha/2)^{th}$ and $(\alpha/2)^{th}$ quantiles from a χ^2 -distribution with $(n-1)$ d.f., respectively.
 - Under the assumption: Population is Normal
- Example 6.4.1; (Page 317)

Estimating population proportion (p)

- Pivotal quantity (Point Estimate): Sample proportion ($\hat{p}$)
- Intervals:
 - $\left[\hat{p} - z_{1-\frac{\alpha}{2}} \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}}, \hat{p} + z_{1-\frac{\alpha}{2}} \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}} \right]$, where $z_{1-\frac{\alpha}{2}}$ is the $(1-\alpha/2)^{th}$ quantile from the standard normal distribution
 - Under the assumption: Sample size (n) is large
Both $np > 5$ and $n(1-p) > 5$
- Example 6.2.4; (Page 303)

Interval Estimation VI

- Width of a $(1-\alpha)100\%$ CI, for the true proportion (p)

$$b = 2z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

- Note: - $b = 2z_{1-\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq z_{1-\frac{\alpha}{2}} \sqrt{\frac{1}{n}}.$
- Margin of error at $(1-\alpha)100\%$ CI, for the true proportion (p)
100d%,

where

$$d = \frac{\max b}{2} = \frac{z_{1-\frac{\alpha}{2}}}{2\sqrt{n}}$$

- Note: - Width and margin of error reduce as sample size increases.

Sample size selection without pilot study

- To estimate p at level $(1 - \alpha)$ to within d (given) units of its true value

- $\frac{z_{1-\frac{\alpha}{2}}}{2\sqrt{n}} \leq d \Rightarrow n \geq \frac{z_{1-\frac{\alpha}{2}}^2}{4d^2}$

- Thus,

$$n = \left\lceil \frac{z_{1-\frac{\alpha}{2}}^2}{4d^2} \right\rceil$$

Sample size selection after pilot study

- Sometimes, we may have an initial estimate $\tilde{p}$ of the parameter p from a pilot study or simulation.
- In this case, $d = \frac{b}{2} = z_{1-\frac{\alpha}{2}} \sqrt{\frac{\tilde{p}(1-\tilde{p})}{n}}$ and

$$n = \left\lceil \frac{z_{1-\frac{\alpha}{2}}^2 \tilde{p}(1-\tilde{p})}{d^2} \right\rceil.$$

Interval Estimation IX

Example:

- Suppose that a local TV station in a city wants to conduct a survey to estimate support for the president's policies on economy within 3% error with 95% confidence.
 - The number of people should be surveyed by the station, if they have no information on the support level is

$$n = \left\lceil \frac{z_{1-\frac{\alpha}{2}}^2}{4d^2} \right\rceil = \left\lceil \frac{1.96^2}{4 \times 0.03^2} \right\rceil = 1068$$

(b) Suppose they have an initial estimate that 70% of the people in the city support the economic policies of the president. Then, the number of people should be surveyed by the station is

$$n = \left\lceil \frac{z_{1-\frac{\alpha}{2}}^2 \tilde{p}(1 - \tilde{p})}{d^2} \right\rceil = \left\lceil \frac{1.96^2 \times 0.7 \times (1 - 0.7)}{0.03^2} \right\rceil = 897$$

Interval Estimation X

Interval Estimation

To Estimate	Assumptions	Interval
Mean	Population is Normal Variance (σ^2) is known	$\left[\bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right]$
	Population is Normal Variance is unknown	$\left[\bar{x} + t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}, \bar{x} - t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}} \right]$
	Sample size (n) is large	$\left[\bar{x} + z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}, \bar{x} - z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \right]$
Proportion	$np > 5$ as well as $n(1 - p) > 5$	$\left[\hat{p} + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right]$
Variance	Population is Normal	$\left[\frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}}, \frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2}, n-1}} \right]$

Interval Estimation XI

- $(1 - \alpha) \times 100$: Confidence level in percentage
- $z_{\frac{\alpha}{2}}$: $\alpha/2^{th}$ quantile from standard normal distribution

$$z_{\frac{\alpha}{2}} = qnorm\left(\frac{\alpha}{2}, mean = 0, sd = 1\right)$$

- $t_{\frac{\alpha}{2}, n-1}$: $\alpha/2^{th}$ quantile from t -distribution with $n-1$ d.f.

$$t_{\frac{\alpha}{2}, n-1} = qt\left(\frac{\alpha}{2}, df = n - 1\right)$$

- $\chi^2_{\frac{\alpha}{2}, n-1}$: $\alpha/2^{th}$ quantile from χ^2 -distribution with $n-1$ d.f.

$$\chi^2_{\frac{\alpha}{2}, n-1} = qchisq\left(\frac{\alpha}{2}, df = n - 1\right)$$

- $\chi^2_{1-\frac{\alpha}{2}, n-1}$: $(1 - \alpha/2)^{th}$ quantile from χ^2 -distribution with $n-1$ d.f.

$$\chi^2_{1-\frac{\alpha}{2}, n-1} = qchisq\left(1 - \frac{\alpha}{2}, df = n - 1\right)$$